Agentic World Modeling: 42 Researchers Map 400+ Papers into a Unified Taxonomy

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A 42-author team published Agentic World Modeling, a survey synthesizing over 400 papers into a single framework for understanding how AI agents model their environments. The paper introduces a "Levels × Laws" taxonomy on two axes. Three capability levels: L1 Predictor (learns one-step transitions), L2 Simulator (composes multi-step, action-conditioned rollouts that respect domain constraints), and L3 Evolver (autonomously revises its own model when

predictions contradict new evidence). Four governing-law regimes: physical (robotics, autonomous driving), digital (code execution, web navigation), social (multi-agent coordination, beliefs and norms), and scientific (hypothesis-driven experimentation, drug discovery). The survey maps 100+ representative systems into this 3×4 grid, spanning model-based reinforcement learning, video generation, web agents, social simulation, and autonomous scientific discovery.

Researchers building AI agents currently pull from at least five disconnected communities: robotics, game AI, web automation, social simulation, and scientific discovery. Each uses incompatible definitions of "world model." This taxonomy gives them a shared coordinate system. If you're building a web agent, you now know you need L2 capability under digital laws, and you can look at what physical-world systems already solved for similar constraints. The cross-domain mapping also reveals which cells are well-explored (L1 physical) and which are nearly empty (L3 social), pointing where new research should go. For related work on how world models handle consistency, see WildWorld.

The framing suggests capable AI agents won't rely on a single model. They'll need a layered stack: reactive prediction for moment-to-moment control, simulation for planning, and self-revision when the environment shifts beneath them.

Sources:

- Agentic World Modeling (arXiv)
- DreamerV3: Mastering Diverse Control Tasks (Nature 2025)
- Generative Agents: Interactive Simulacra (Park et al. 2023)
- LeCun: A Path Towards Autonomous Machine Intelligence

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